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SUBJECT CODE NO: - RNP-8053
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Mechanical) (NEP) Sem - IV
Examination May/June-2026
Strength of Materials

[Time: 1:30 Hours]**[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from remaining Part B: Q.2 to Q.7

Part A**Q.1 Solve any 5 following questions****10**

- a) What is bulk modulus (K)?
- b) What is ultimate stress?
- c) What is a flitched beam?
- d) What is polar moment of inertia (J)?
- e) What is proof resilience?
- f) What is factor of safety?

Part B**Solve any four from the following**

- | | |
|--|----------|
| Q.2 Explain Poisson's ratio and derive relation between elastic constants. | 5 |
| Q.3 Write short note on types of beams and loads. | 5 |
| Q.4 A cantilever beam of uniform c/s section $200 \times 300\text{mm}$ rectangular, carries U.d.L of 20 KN/m over its span 2m . Determine the maxim bending stress anywhere in the beam | 5 |
| Q.5 Explain torsion in solid and hollow shafts. | 5 |
| Q.6 Derive expressions for principal stresses and maximum shear stress. | 5 |
| Q.7 Derive Euler's formula for columns. | 5 |